

Does the AR-HUD system affect driving behaviour? An eye-tracking experiment study

Yu-nuo Cheng^a, Xia Zhong^{a,*}, Li-wei Tian^b

^a School of Architecture and Design, China University of Mining and Technology, Xuzhou, Jinagsu 221116, China

^b Intelligent Research Institute, Changan Automobile Group Co., Ltd., Chongqing 401133, China

ARTICLE INFO

Keywords:

AR-HUD
Driving behaviour
Risk driving
Eye movement
Area of interest

ABSTRACT

In recent years, the continuous development of in-vehicle information systems (IVIS) has greatly enriched the driving experience, while also occupying the driver's cognitive resources to varying degrees, causing driving distraction. Therefore, studying the influence of the AR-HUD (Augmented Reality Head-up Display, AR-HUD) system on driving behaviour is of great significance for the in-vehicle information system to manage the complexity of information and enhance driving safety. This experiment applied five typical risky driving scenarios under two lighting conditions (daytime and night), as stimulus materials. The effects of eye movement behaviour and risk reaction time indicators during driving were comprehensively analysed. The experiment results showed that the AR-HUD system can significantly improve the subjects' attention to risky AOIs (Area of Interest, AOI) in night driving situations, as well as reduce the difficulty of processing information in risky driving scenarios, thus reducing cognitive load; In terms of reaction time, the AR-HUD system can significantly reduce the driver's perception time for risky driving scenarios, thus responding to risky situations more quickly. The experiment conclusions verified the role of AR-HUD technology in improving driving safety and driving behaviour, and provide a new direction for the future development of in-vehicle information systems.

1. Introduction

Information technology, represented by the Internet and communication technology, is markedly affecting the social foundation and life forms of human beings (Venkatesh, 2003). As an important field and direction to penetrate the mobile Internet, automobiles will become the main field of digitisation in the future. A large number of in-vehicle information systems, such as mobile phones, satellite radios and in-vehicle applications are constantly pouring into the car, and the IVIS has developed from the early basic driving information to a multidimensional interaction system (Hao et al., 2015; Liang, 2007). While under this system, some driving tasks that are not directly related to driving or even irrelevant are occupying the driver's cognitive resources to varying degrees, causing driving distraction (Patten, 2004; Dong et al., 2010). According to accident statistics from the National Highway Traffic Safety Administration (NHTSA), >18% of injuries and 9% of fatalities in 2010 were caused by driver distraction (Nhtsa, 2012). During driving, besides completing the main driving task, the drivers also perform many things that are irrelevant to the main tasks (Ma and

Kaber, 2005; Liu, 2017). However, with limited cognitive resources, it is difficult for drivers to allocate their attention effectively among tasks (Cooper et al., 2013; Castro, 2019; Ma, 2020). Therefore, in such a complex information system, how to manage the complexity of the IVIS effectively and further improve driving safety is a key problem that needs to solve urgently in the current IVIS.

The all-over display represents an important symbol in digitalising and informatising of the future automobile. Particularly, 80 percent of the driving information is visually perceived (Venkatasawmy, 2020), so visual support is the top priority of the vehicle assistance driving system. Head-up Display (HUD), as a visual assisted technology, has attracted extensive attention in recent decades (Xing and Qi, 2014). From the perspective of the display technique, HUD will gradually substitute part of the functions of the instrument panel to minimize the deviation of the driver's sight from the road ahead due to looking down at the instrument (Liu and Wen, 2008; Medenica, 2011). The content of the HUD system is primarily based on driving information. In addition, some secondary task information, such as music, phone reminders, etc., will also display in the non-important area of HUD. The emergence of HUD is conducive

Abbreviations: AR-HUD, Augmented Reality Head-up Display; FOV, Field of View; IVIS, In-vehicle Information System; AOI, Area of Interest.

* Corresponding author.

E-mail address: 873426189@qq.com (X. Zhong).

<https://doi.org/10.1016/j.trp.2023.100767>

Received 28 September 2022; Received in revised form 3 January 2023; Accepted 29 January 2023

Available online 3 February 2023

2590-1982/© 2023 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

to reducing the driver's distraction caused by reading vehicle information while driving (Ward and Parkes, 1994; Grandi et al., 2022; Choi et al., 2002). Since the HUD is placed in the driver's front field of view (FFOV), it is favourable to reduce the occupation of visual, motive and cognitive resources. According to references (Gibbered et al., 2010; Wang et al., 2016; Bellazzi et al., 2022), head-up display technology can decrease vehicle crashes by 25%. Augmented reality (AR) is a new type of human-computer interaction technology based on virtual reality (Fan and Liang, 2012; Fröhlich, 2010). Combining AR technology and HUD enables vehicles to carry both natural and artificial displays at the same time. It superimposes virtual information such as road conditions, navigation, and non-geometric information of real objects (e.g. pedestrians crossing the road) to real driving scenes (Smith et al., 2017; Lee et al., 2009). Due to its characteristics of combining virtual and real, and real-time interaction, it has played a strong driving role in the development of future vehicle assisted driving in-vehicle entertainment and navigation (Choi et al., 2013; Hwang et al., 2015).

In recent years, the implementation of AR technology in IVIS has become a hot research topic for major automobile companies. Experts and scholars have conducted a lot of experiments on driver assistance systems based on AR and HUD technologies, and have achieved many results. Park (Park and Kim, 2014) developed a vehicle-mounted AR-HUD system that can recognize driving safety information and project it into the driver's FOV. The recognition rate of driving safety information can reach 73%, and the recognition speed can achieve 15 fps. Changrak Yoon et al. (Yoon et al., 2014) proposed a pre-collision warning system that combines AR technology and head-up display. The system detects vehicles and pedestrians ahead, judges dangerous driving situations based on the detection results, and uses AR technology demonstrations to warn drivers. Bark K (Bark et al., 2015) et al. proposed an AR-HUD navigation assistance system design interface. By comparing the driver's reaction time in two situations with and without AR, the experiment result showed the AR-HUD navigation system can help the driver respond to the turning position faster. Lee JW (Lee et al., 2015) et al. summarized the problems of vehicle AR-HUD systems in display equipment, target recognition equipment, information organization and visual design methods, and experimental methods. Kim H et al. (Kim et al., 2016) used HUD to design a new pedestrian detection interface, and the virtual shadow of the pedestrian was covered by the front windshield in the driver's FOV. The results showed that applying the system interface resulted in better driving performance and smoother braking behaviour. Hwang Y et al. (Hwang et al., 2016) tested the impact of the AR-HUD system on the driver's risky-response ability and the driver's psychological change. Kim K (Kim and Hwang, 2017) verified through experiments whether the AR-HUD system can effectively improve driving safety and whether the visual symbols covered in the real driving environment will interfere with the driving vision. Zhang Huawei (Huawei, 2015) used various methods of ergonomics and design psychology to explore the design of the visual interface of the head-up display system to improve the driver's cognitive load and driving safety.

A comprehensive analysis of the research status found that there are two main aspects to the application of AR and HUD technology to IVIS: ① Apply AR and HUD technology to the forward-collision warning system, project the detected information of obstacles such as pedestrians and vehicles on the front windshield, and combine with the real driving environment to improve the driver's situation awareness of risky driving ability; ② Apply AR and HUD technologies to the vehicle navigation system, and the visual navigation information is directly covered in the road environment to improve the intuitive and easy-to-read navigation system. Most of the research experiments focus on evaluating the effectiveness of the proposed system prototype by comparing the driving performance in different driving situations. However, there is a lack of further discussion on the impact of the AR-HUD system on improving driving distraction and the cognitive and decision-making process in dealing with risky driving situations. As a consequence, the main

purpose of this paper is to explore the influence of the AR-HUD system on driving behaviour. In this paper, a simulated driving experiment was conducted to investigate the impact of the AR-HUD system on driving behaviour, and we demonstrated a comparison between the AR-HUD group and the control group (without AR-HUD). The results of this comparison showed that AR-HUD technology plays a vital role in enhancing driving safety and driving behaviour, which is conducive to solving the current problems faced by IVIS, and provided a new direction for the future interface design of the AR-HUD system.

2. Method

2.1. Participants

Due to differences in gender, age, driving experience and operating habits, etc., the acceptance of the AR-HUD system is different correspondingly. In general, younger drivers are more interested in using intelligent assisted driving systems, while older drivers are more concerned about driving safety and stability. At the same time, the driver's psychological condition will also affect their driving behaviours.

The subjects were 30 drivers who volunteered to participate in Changan Automobile R&D Centre, including 20 males and 10 females, aged 21–52 years old, including novice drivers and professional test drivers, as shown in Table 1. All assigned subjects were in good health and mental state, vision (including corrected) is normal, and no achromatopsia or tritanopia. This experiment was authorized by the Safety Department and the Intelligent Research Institute of Changan Automobile Company.

The subjects were assigned randomly to the AR-HUD group and the control group. A one-way analysis of variance (ANOVA) result showed there was no statistically significant difference between the two groups.

2.2. Apparatus

This experimental apparatus consisted of a driver-centric driving environment through an intelligent car prototype (unpowered system), which was equipped with an AR-HUD system, and its FOV is 12×5 . The simulated driving scenarios are simulated and controlled by SCANer Studio, as shown in Fig. 1. In addition, the recording system for eye movements was the Tobii Pro Nano (Tobii Technology). This eye-tracking system has a sampling rate of 60 Hz that can accurately track the location of the focus of both eyes (maximal time threshold is 75 ms, and the maximum angle threshold is 0.5), as shown in Fig. 2. The driving environment is portrayed through a large curved projection screen, which maximises the realization of a vehicle simulation environment similar to the real driving environment.

2.3. Stimuli

To corroborate the impact of the AR-HUD system on driving behaviour in different driving situations, the five principles for developing driver risky perception test proposed by Mark A (Mark and Wetton, 2011), the five most typical risky driving scenarios were extracted, including pedestrian conflict (S1), vehicle ahead slows down (S2), vehicle turns at an intersection (S3), ramp merge (S4) and expressway

Table 1
Basic information of participants.

	Age		Driving experience/ (year)		Accumulated miles driven (/100 km)	
	M _{age}	SD _{age}	M _{exp.}	SD _{exp.}	M _{miles}	SD _{miles}
AR-HUD	30.60	9.27	4.72	2.88	5.31	8.25
Control group	31.00	9.18	5.13	3.51	6.02	7.42

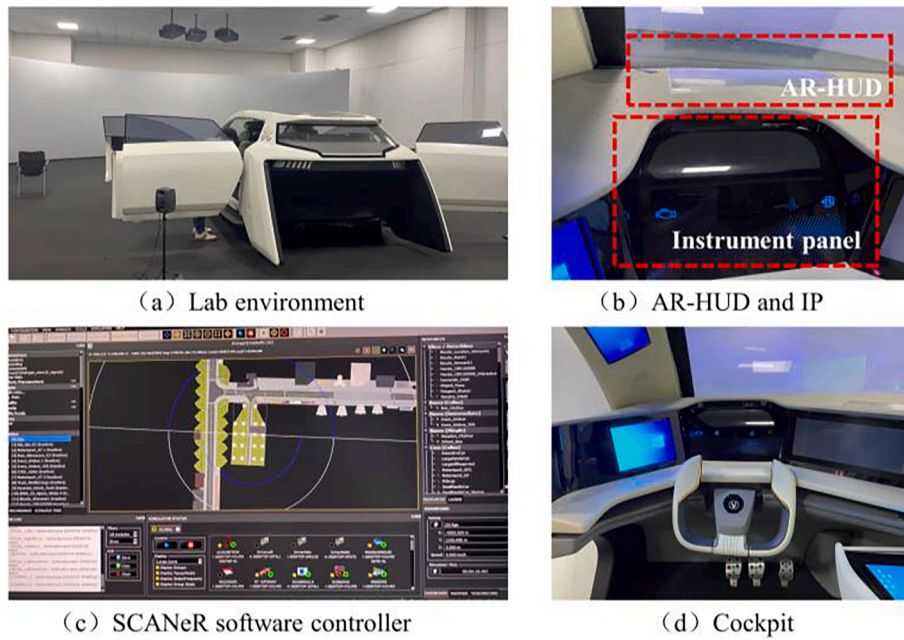


Fig. 1. Vehicle simulation environment.

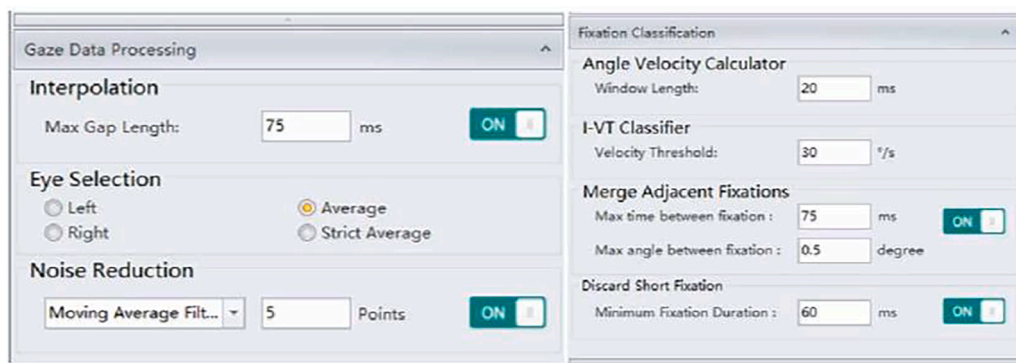


Fig. 2. Eye tracker parameter settings.

overtaking (S5). These five typical risky driving scenarios are interspersed in the fixed driving route respectively, forming two driving conditions, i.e. daytime and night.

2.4. Experiment design

We performed a 2×5 factorial design for the driving behaviour study. The independent variables are the AR-HUD group and control group (without AR-HUD), and the dependent variables are fixation behaviour and risky-response time of five AOIs in different scenarios. In this experiment, the count of fixations, mean fixation time, and fixation duration percentage were selected as indicators of the driver's fixation behaviour. Correspondingly, the risky-perception time and risky decision-making time were selected as indicators of the driver's risky-response capability.

2.4.1. Fixation behaviour in AOIs

The dynamic area of interest (AOI) needs to be drawn to analyse the area where the risky driving scenario occurs separately. The shape and transformation of the AOI need to be defined based on keyframes, as shown in Fig. 3. The driving simulation programme in this experiment contains five potentially risky driving scenarios, thus creating five dynamic AOIs in each scenario.

2.4.2. Risky-response time

The risky-response time is divided into two parts: risk perception time and risk decision-making time. Risk perception time refers to the time difference between the appearance of the risk and participants detecting the risk. And risk decision-making time refers to the time when participants provide feedback after detecting the risk, as shown in Fig. 4.

2.5. Procedure

The duration of the experiment was about 30 min. Participants were asked to read and accept a confidential declaration of consent. Afterwards, participants were divided into two groups randomly (AR-HUD group and control group), which is not statistically different. Then we introduced the experimental purpose, the procedure and the precautions to the two groups. And introduced the design conception, main content and the meaning of each visual symbol of the AR-HUD system to the AR-HUD group. Before the experiment began, subjects in the AR-HUD group were reminded to fill in a subjective evaluation scale (pre-evaluation) to evaluate their first impression of the system interface and were required to complete the five-point calibration verification and the gaze accuracy test.

After entering the vehicle and adjusting the seat, 5 min of free driving practice was allowed. During the driving practice, participants

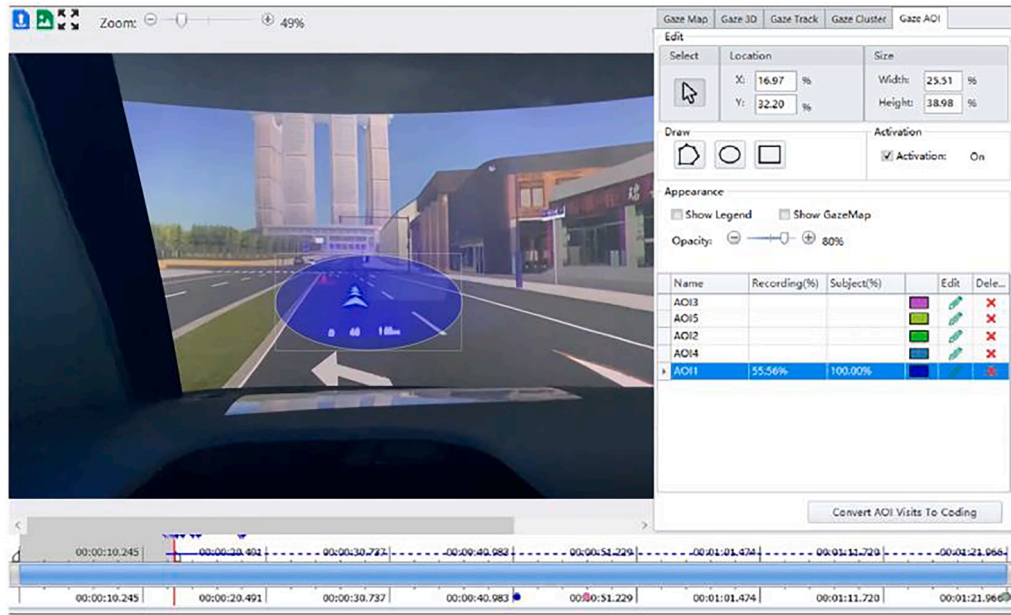


Fig. 3. Dynamic AOI rendering in S1.

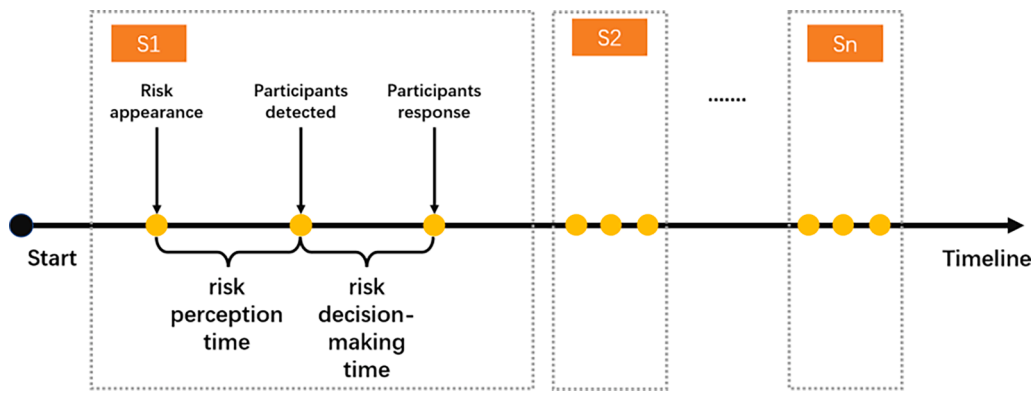


Fig. 4. The definition of risky-response time.

had to drive manually and familiarise the surrounding environment. In the formal experiment, when subjects drove into a certain segment, a stimulus appeared, e.g. Pedestrian conflict. Subjects needed to detect the stimulus as soon as possible and press the button on the steering wheel, and then subjects needed to manoeuvre the vehicle to avoid pedestrians. These three time-points (stimulus appearance, risk detecting and button pressing) were recorded by the system automatically. After finishing the experiment, the AR-HUD group were asked to refill the evaluation scale again(post-evaluation), as shown in [Appendix A](#). The next participant continued to experiment. [Fig. 5](#) shows the whole procedure of the experiment.

3. Results

Based on the experimental data collected by the eye tracker, the impact of the AR-HUD system on the driver's driving behaviour is analysed from three perspectives: eye movement data, the risky reaction time and subjective evaluation. SPSSAU 2022 ([The SPSSAU project, 2022](#)) was applied for the statistics and analysis of experimental data.

3.1. Fixation behaviour

3.1.1. Fixation counts

[Table 2](#) shows the mean fixation counts of the five AOIs of the two groups. [Fig. 6](#) shows the subjects' fixation counts boxplot on the AOI in two driving situations during daytime and night.

In the daytime driving, results showed the mean count of fixation of the subjects in the AR-HUD group is decreased by 5.82%, and the SD is increased by 6.59% compared with the control group; the boxplot showed that the mean count of fixation in the AR-HUD group levels was slightly lower than those in the control group. In the night driving situation, results showed that the mean count of fixation in the AR-HUD group is 22.04% higher than that of the control group, and the SD is increased by 1.94%; the boxplot showed the mean count of fixation in the AR-HUD group was significantly higher than the control group, and the box height IQR(interquartile range, IQR) was significantly higher than that of the control group, i.e., the data fluctuation degree of the AR-HUD group was significantly greater than that of the control group.

The paired sample T-test results of the fixation count data are shown in [Table 3](#). In the daytime driving situation, $p = 0.226 > 0.05$, which means the effect of the driver's fixation frequency in the driving situation is not significant; In the night driving situation, $p = 0.019 < 0.05$, which is significant, i.e., the use of the AR-HUD can significantly boost

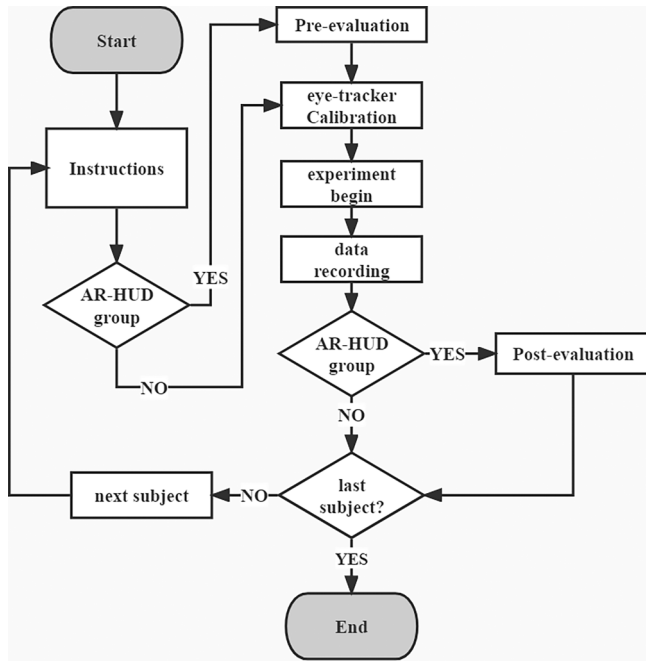


Fig. 5. Experiment procedure.

Table 2
Average fixation statistics(count).

		S1	S2	S3	S4	S5	M	SD
Daytime	AR-HUD	3.95	2.33	2.37	6.67	3.27	3.72	1.78
	Control group	4.52	2.92	2.26	6.53	3.50	3.94	1.67
Night	AR-HUD	7.15	4.86	4.26	7.93	5.13	5.87	1.58
	Control group	5.80	3.13	4.21	6.98	3.93	4.81	1.55

the driver's attention to the AOI in the night driving situation.

3.1.2. Mean fixation time

Table 4 shows the mean fixation time of the five AOIs in two groups, and Fig. 7 shows the mean fixation time of the subjects to the AOI in two driving situations during the daytime and night.

In the daytime driving situation, results showed that the mean

fixation time of the AR-HUD group was reduced by 2.88%, and the standard deviation is increased by 10.03% compared with the control group; The boxplot shows that the mean fixation time of the AR-HUD group is lower than that of the control group, and the box height IQR is significantly higher than that of the control group. In the night driving situation, results showed that the mean fixation time of the AR-HUD group was reduced by 10.00%, and the standard deviation was increased by 49.34% compared with the control group; The boxplot shows that the mean fixation time of the subjects in the AR-HUD group was significantly lower than that of the control group, and the box height IQR was significantly higher than that of the control group.

The paired sample T-test results of the mean fixation time are shown in Table 5. In the daytime driving situation, $p = 0.170 > 0.05$, which is not significant, even if the AR-HUD assisted driving system is implemented, the effect on the mean fixation time of the driver in the daytime driving situation is not significant; In the night driving situation, $p = 0.009 < 0.05$, which is significant, i.e. the use of the AR-HUD system can significantly reduce the driver's cognitive load on the risky driving situation in the night driving situation.

3.1.3. Fixation duration percentage

Table 6 shows the fixation duration percentage for the AOIs and the statistical results of the two groups. Fig. 8 shows the boxplot of the fixation duration percentage for the subjects to fixate on the AOI in two driving situations during the daytime and night.

In the daytime driving situation, results show that the mean fixation duration percentage of the AR-HUD group is 9.47% lower than that of the control group, and the standard deviation is increased by 6.85%. The boxplot shows that the mean fixation duration percentage of the AR-HUD group is slightly lower than that of the control group, but the box height IQR is significantly higher than that of the control group. In the night driving situation, results show that the AR-HUD group's mean fixation duration percentage, is 14.35% higher than that of the control group, and the standard deviation is 58.92% higher. The boxplot shows that the mean fixation duration percentage and the box height IQR of the subjects in the AR-HUD group were significantly higher than those in the control group.

3.2. Risky-response capability

The time from the appearance of the danger signal to the time when the subject pressed the button was defined as the risky-response time, i.e., the perception-response reaction time. In this experiment, the risky-response behaviour is divided into risky perception time and risky decision-making time.

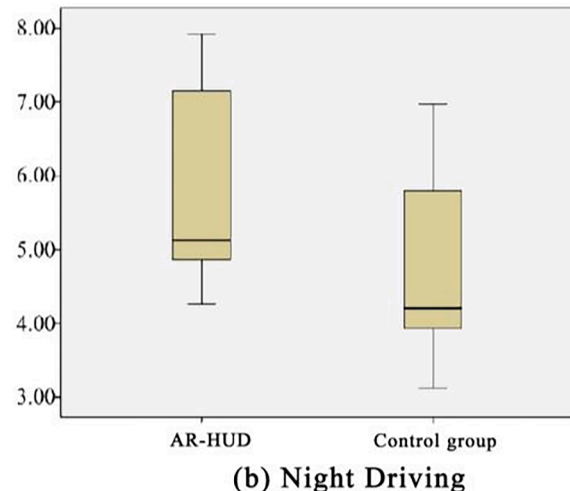
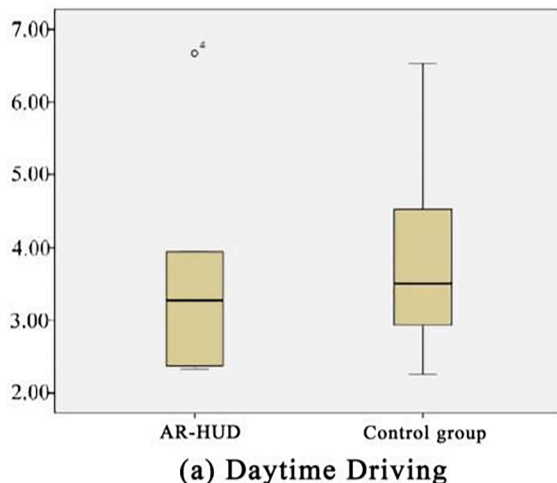


Fig. 6. Average fixation count.

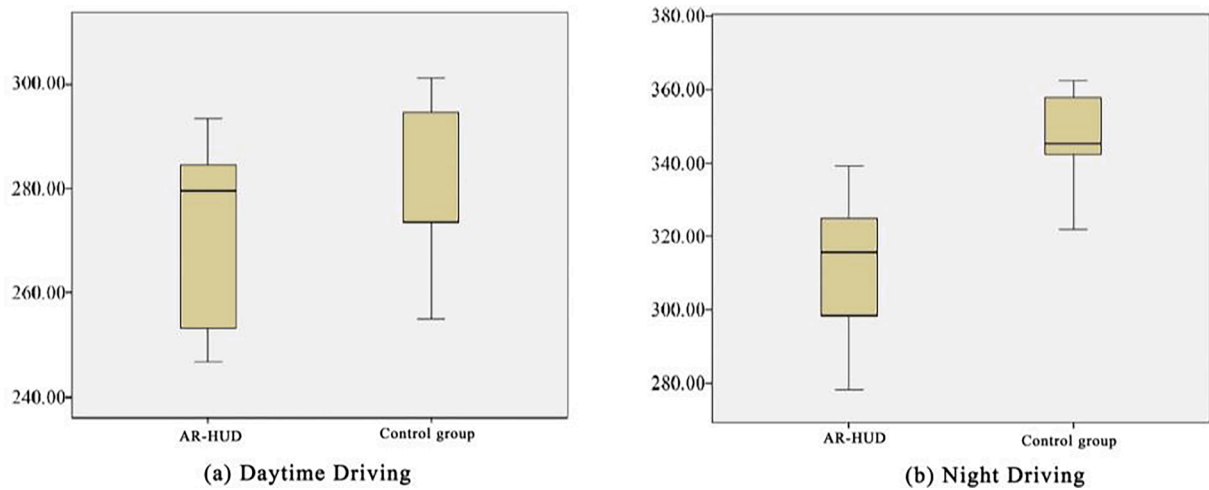
Table 3Paired sample *t*-test for fixation count.

Test		Coupled difference				t	df	Sig. (2-tailed)	
		Mean	SD	SE	95% confidence interval				
					lower limit				upper limit
Fixation count	Daytime	−0.232	0.363	0.162	−0.681	0.218	−1.431	4	0.226
	Night	1.058	0.625	0.280	0.281	1.835	3.782	4	0.019*

Table 4

Average fixation time(ms).

		S1	S2	S3	S4	S5	Mean	SD
Daytime	AR-HUD	246.71	293.44	284.57	253.27	279.48	271.50	20.40
	Control group	254.93	294.58	301.21	273.45	273.54	279.54	18.54
Night	AR-HUD	298.45	315.78	324.96	278.41	339.27	311.37	23.64
	Control group	342.36	357.92	362.36	321.94	345.30	345.97	15.83

**Fig. 7.** Average fixation time.**Table 5**Paired sample *t*-test for the mean fixation time.

Test		Coupled difference				t	df	Sig. (2-tailed)	
		Mean	SD	SE	95% confidence interval				
					lower limit				upper limit
Mean fixation time	Daytime	−8.044	10.778	4.820	−21.427	5.339	−1.669	4	0.170
	Night	−34.600	16.186	7.239	−54.697	−14.503	−4.780	4	0.009*

Table 6

Statistical results of the percentage of fixation time (%).

		S1	S2	S3	S4	S5	Mean	SD
Daytime	AR-HUD	30.99	22.17	27.97	30.26	23.85	27.05	3.90
	Control group	35.58	27.38	28.71	31.22	26.49	29.88	3.65
Night	AR-HUD	45.72	32.75	35.79	42.38	30.28	37.38	6.50
	Control group	38.21	30.95	30.27	35.66	28.36	32.68	4.09

3.2.1. Perception time

In the ErgoLAB system, dynamic AOIs were plotted according to the event, location and period of the danger signal. Among them, the moment dynamic AOI starts to be drawn is defined as the moment the danger signal appears. And the moment subjects first entered the AOI recorded by the ErgoLAB system is defined as the moment subjects detected the signal. The specific data are shown in Table 7.

As shown in Table 8 and Fig. 9, the results show that in the daytime driving situation, the AR-HUD group's perception time in various risky driving situations is shortened by 17.91% compared with the control group, and the standard deviation is increased by 20%. The boxplot shows that the average level of risky perception time of the AR-HUD group is slightly lower than that of the control group, and the box height IQR is significantly lower than that of the control group. In the

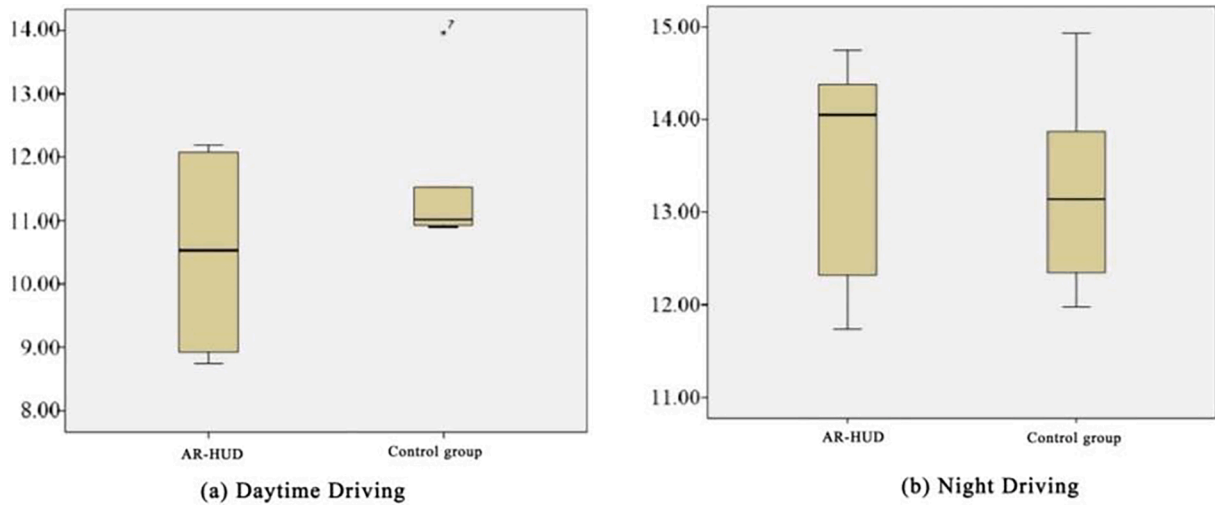


Fig. 8. Fixation duration percentage.

Table 7

Risk perception time data (s).

		Risk appears		Risk detected		Perception time	
		AR-HUD	Control group	AR-HUD	Control group	AR-HUD	Control group
Daytime	S1	12.34	14.53	12.97	15.36	0.63	0.83
	S2	34.12	34.60	34.39	35.07	0.27	0.47
	S3	35.95	39.88	36.71	40.69	0.76	0.81
	S4	58.39	59.68	58.91	60.33	0.52	0.65
	S5	63.94	70.95	64.53	71.55	0.59	0.60
Night	S1	10.22	12.47	11.12	13.49	0.90	1.02
	S2	18.24	18.57	19.18	19.72	0.94	1.15
	S3	44.43	45.32	45.30	46.37	0.87	1.05
	S4	54.96	55.67	55.84	56.97	0.84	1.30
	S5	75.79	70.47	76.50	71.29	0.71	0.82

Table 8

Risky perception time statistical results (s).

		Mean	SD	Minimum	Maximum	Midian
Daytime	AR-HUD	0.55	0.18	0.27	0.76	0.59
	Control group	0.67	0.15	0.47	0.83	0.65
Night	AR-HUD	0.85	0.09	0.71	0.94	0.87
	Control group	1.07	0.18	0.82	1.30	1.05

night driving situation, the AR-HUD group's perception time in various risky driving situations is shortened by 20.56% compared with the control group, and the standard deviation I reduced by 50%. The boxplot shows that the average level of risk perception time and the IQR of the box height of the AR-HUD group are significantly lower than those of the control group.

The results of the paired samples T-test are shown in Table 9. When the significance level is 0.05, in the daytime driving situation, the daytime driving situation $p = 0.038 < 0.05$, the effect is significant; In

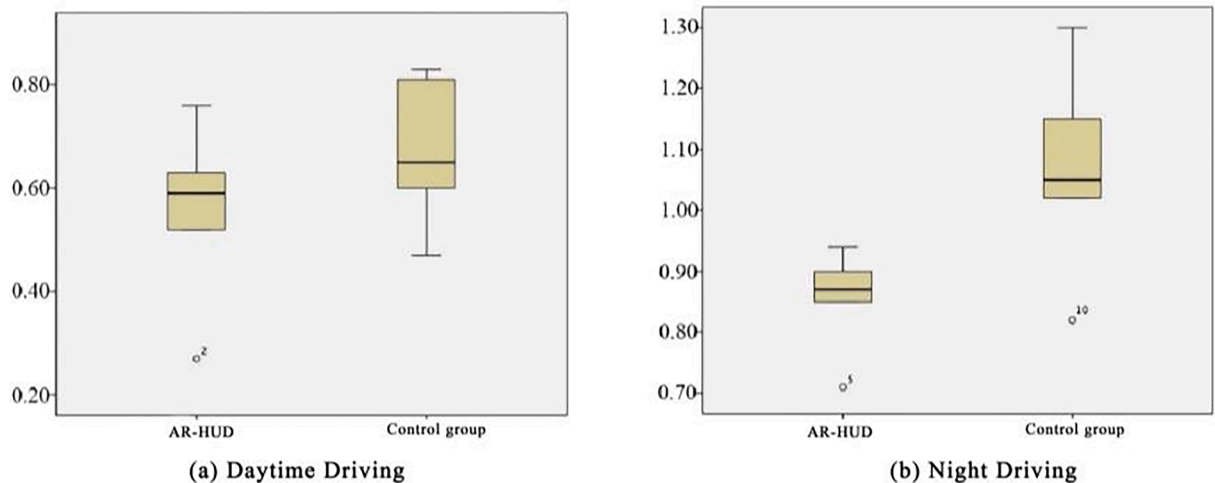


Fig. 9. Risky Perception Time.

the night driving situation, $p = 0.026 < 0.05$, the effect is significant. i.e. the implementation of the AR-HUD system can significantly shorten the perception time of the subjects in risky driving situations.

3.2.2. Decision-making time

The risky decision-making time in this experiment was defined as the time from subjects detected the danger signal to the time they responded. Among them, the time when the subject pressed the button was defined as the subject's reaction time. Data are shown in Table 10.

The decision-making time analysis is shown in Table 11, and the boxplot is shown in Fig. 10. Results showed that in the daytime driving situation, the AR-HUD group's decision-making time in various risky driving situations was shortened by 14.04% compared with the control group, and the standard deviation was reduced by 55.56%; In the night situation, the driver's decision-making time in various risky driving situations in the AR-HUD group was reduced by 7.81% compared with the control group, and the standard deviation was reduced by 9.09%. The boxplot showed that in the daytime and night driving situations, the AR-HUD group's average level of risky decision-making time was lower than that of the control group, and the box height IQR was significantly higher than that of the control group.

A paired sample T-test was performed and the results are shown in Table 12. In the daytime driving situation, $p = 0.064 > 0.05$, the effect was not significant; in the night driving situation, $p = 0.164 > 0.05$, the effect was not significant, i.e. using the AR-HUD system has no significant effect on the perception time of the subjects' risky driving situations.

3.3. Subjective evaluation

To obtain the driver's subjective evaluation of the effectiveness, interference and reliability of the AR-HUD system, 15 subjects in the AR-HUD group were asked to complete the subjective evaluation scales before and after the experiment. 15 valid evaluation scales were collected, and the data are shown in Table 13.

Fig. 11(a) illustrates the comparison of the subjective evaluation of the AR-HUD system in different experiment stages. It can be seen that after implementing the AR-HUD system, the effectiveness evaluation of the subjects had been improved than expected, worries about the system obscuring the driving environment, interfering with normal driving, and distracting the driver had lessened than expected as well. i.e. After using the system, the subjective acceptance of the AR-HUD system improved.

In the AR-HUD group, there are 7 novices and 8 skilled drivers. The evaluation scale data of subjects with different driving experiences are analysed separately, and the comparison chart is shown in Fig. 11(b); For subjects with different driving experiences, the novice's evaluation of the system's legibility was higher, and the score of adding pre-collision warning to the system and distinguishing it by different colours and frequencies was slightly lower. The main reason is that various warning forms are easy to cause confusion and distract the drivers (Jenness and Singer, 2073.104-113.2008.). Comprehensive analysis showed that skilled drivers were more acceptable to the system, believing that the system makes the driving process easier, while novice drivers were more sceptical about the system, considering that the AR-HUD system makes drivers more nervous.

There were 10 male drivers and 5 female drivers in the AR-HUD

group. The evaluation scale data of subjects of different genders are shown in Fig. 11(c). It was found that female drivers scored higher on the interference of the system, concerning the system occluded the ambient environment and distracted the FOV. In Smith's research (Smith, 2010), it was pointed out that the location of the HUD interface is an important factor affecting the reliability of the system for females. i.e. If the image was projected on the right side of the steering wheel, they felt "relaxing", but if the image was projected on the left side, they felt "anxious". And female drivers relied more on the system and were more inclined to use the AR-HUD system to judge driving risks.

4. Discussion

In this study, we compared the effects of AR-HUD and the control group on driving behaviour under 5 driving scenarios, especially the driver's eye movement behaviour, i.e. the count of fixations, the mean fixation time and fixation duration percentage. Next, we will discuss each result-related variable separately.

Fixation behaviour includes fixation count, mean fixation time, and fixation duration percentage. The driver's fixation behaviour indicates the acquisition and processing of relevant driving information (Yamaguchi et al., 2019). Generally speaking, the more the count of fixations, the more information is received, and the more frequently the driver observes the area. This indicator mirrors the driver's AOI to a certain extent. In this experiment, the count of fixations is introduced, and the driver's AOI in risky driving situations is analysed by measuring the total count of fixations on the AOI when the driver is in a risky driving situation with (or without) the AR-HUD system. The results showed that the subjects in the AR-HUD group still allocated a small part of their attention to the state information display area of the AR-HUD system when they detected a risky situation during daytime driving, while the attention of the control group was mainly centred over risky AOI. It showed that in the daytime driving situation, the interface information of the AR-HUD system will intervene in the driver's FOV. Therefore, in the design process of the system interface, the legibility and rationality of the AR-HUD system information need to be repeatedly considered (Ward, 1995). While at night, the AR-HUD system helped the driver to concentrate more on the risky area, but its impact varied in different driving scenarios. Thus, it was preliminarily estimated that in the night driving situation, the AR-HUD system can significantly improve the subjects' attention in risky driving areas. Park et al. (Park et al., 2015) also found a similar conclusion that the AR-HUD system can enhance the prompting of hazard signals in night driving and minimize the risk of collision.

The mean fixation time refers to the average duration of each fixation point in the AOI, and this indicator reflects the difficulty for drivers to grasp driving information in the AOI to a certain extent (Lehtonen et al., 2014). In this experiment, the mean fixation time characterization index was introduced to verify whether the AR-HUD system can effectively reduce the driver's cognitive load in risky driving situations. Results showed that in the daytime situation, the AR-HUD system was beneficial to attenuate the difficulty to process the information and reduce the cognitive load. However, the effect is not significant, and the degree of influence differs significantly in different risky driving situations. The main cause may be that external ambient luminance will affect the visibility of AR-HUD, resulting in an inconspicuous display interface

Table 9
Paired sample *t*-test for risky perception time.

Test		Coupled difference					t	df	Sig. (2-tailed)
		Mean	SD	SE	95% confidence interval				
					lower limit	upper limit			
Risky perception time	Daytime	−0.118	0.086	0.039	−0.225	−0.011	−3.053	4	0.038*
	Night	−0.214	0.138	0.062	−0.386	−0.042	−3.460	4	0.026*

Table 10
Risky decision-making time (s).

		Danger detected		Respond moment		Decision-making time	
		AR-HUD	Control group	AR-HUD	Control group	AR-HUD	Control group
Daytime	S1	12.97	15.36	13.48	16.07	0.51	0.71
	S2	34.39	35.07	34.91	35.61	0.52	0.54
	S3	36.71	40.69	37.22	41.25	0.51	0.56
	S4	58.91	60.33	59.34	60.80	0.43	0.47
	S5	64.53	71.55	64.99	72.11	0.46	0.56
Night	S1	11.12	13.49	11.87	14.32	0.75	0.83
	S2	19.18	19.72	19.71	20.29	0.53	0.57
	S3	45.30	46.37	45.87	46.97	0.57	0.60
	S4	55.81	56.97	56.30	57.59	0.49	0.62
	S5	76.50	71.29	77.13	71.88	0.63	0.59

Table 11
Risky decision-making time results (s).

		Mean	SD	Minimum	Maximum	Midian
Daytime	AR-HUD	0.49	0.04	0.43	0.52	0.51
	Control group	0.57	0.09	0.47	0.71	0.56
Night	AR-HUD	0.59	0.10	0.49	0.75	0.57
	Control group	0.64	0.11	0.57	0.83	0.60

(Moffitt, 2019); While in the night situation, the AR-HUD system significantly reduced the difficulty for the driver to process the information of the dangerous driving situation and reduced the cognitive load. In addition, the degree of influence is different in various risky driving situations.

The fixation duration percentage refers to the percentage of the sum of the duration of all fixation points in the AOI to the total time of the AOI (Zhang et al., 2021). Since the duration sum of fixation points reflects the interaction between the count of fixation and the mean fixation time, this characterization index was introduced in this experiment to comprehensively evaluate the effects of the AR-HUD system on driver distraction and cognitive load. The results showed that in the daytime

situation, the AR-HUD system's ability to reduce the driver's driving distraction and cognitive load is not obvious, and the data fluctuates greatly; In the night situation, the AR-HUD system can significantly reduce the driver's driving distraction and cognitive load, and the degree of impact varies significantly in different risky driving situations.

In addition, the risky response capability (risk perception time and risk decision time) reflects the ability to aware and response to the risky situations. As for risk perception time, participants who used the AR-HUD system can obtain risky driving situation information more quickly according to the warning information of the system. And with the help of the system, the risk perception time tended to a certain value in various driving situations, which was more significant in night driving situations. As for risk decision time, AR-HUD system improved the cognitive load of participants in dealing with risky driving information to some extent, and improved the information processing efficiency. However, in different risky driving situations, the degree of influence was obviously different.

The main purpose of this research is to investigate the effect of the AR-HUD system on driving behaviour in 5 different risky scenarios. Nevertheless, this work still has some limitations, for example, the simulation experiment was not carried out on the actual road, ignoring

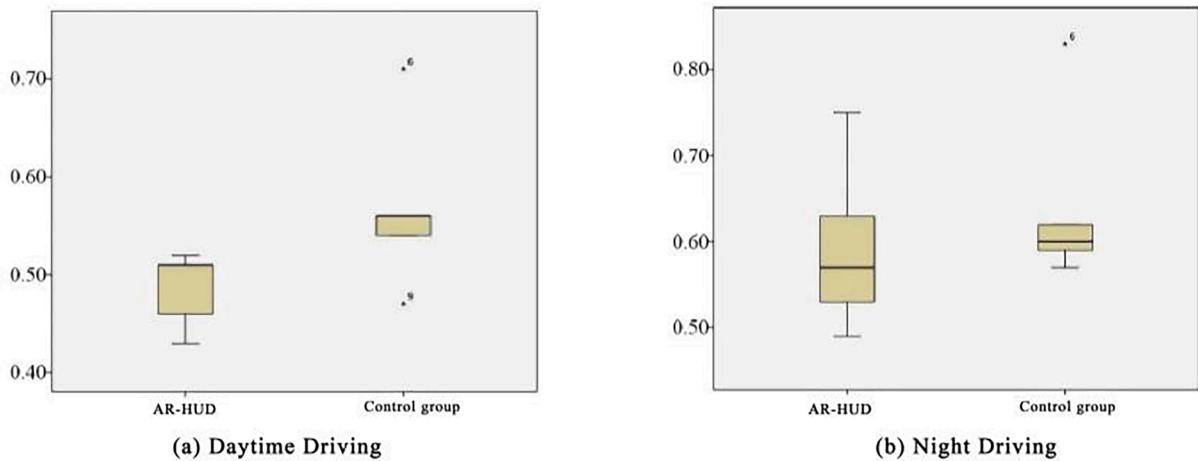


Fig. 10. Risky decision-making time.

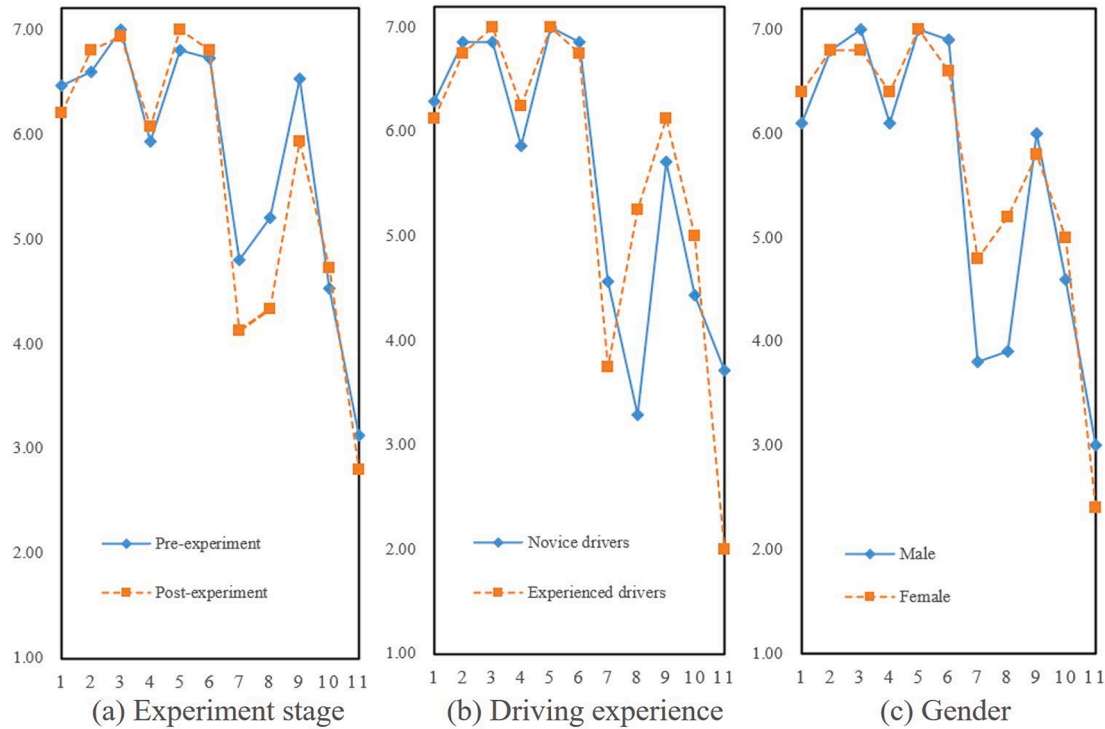
Table 12
Paired sample *t*-test for risky decision-making time.

Test		Coupled difference				t	df	Sig. (2-tailed)	
		Mean	SD	SE	95% confidence interval				
					lower limit				upper limit
Risky decision-making time	Daytime	−0.082	0.072	0.032	−0.172	0.008	−2.538	4	0.064
	Night	−0.048	0.063	0.028	−0.126	0.030	−1.703	4	0.164

Table 13

Statistical results of subjective rating scale data.

Question number	Pre-experiment	Post-experiment	Novice drivers	Experienced drivers	Male	Female
1	6.47	6.20	6.29	6.13	6.10	6.40
2	6.60	6.80	6.86	6.75	6.80	6.80
3	7.00	6.93	6.86	7.00	7.00	6.80
4	5.93	6.07	5.86	6.25	6.10	6.40
5	6.80	7.00	7.00	7.00	7.00	7.00
6	6.73	6.80	6.86	6.75	6.90	6.60
7	4.80	4.13	4.57	3.75	3.80	4.80
8	5.20	4.33	3.29	5.25	3.90	5.20
9	6.53	5.93	5.71	6.13	6.00	5.80
10	4.53	4.73	4.43	5.00	4.60	5.00
11	3.13	2.80	3.71	2.00	3.00	2.40

**Fig. 11.** Subjective evaluation.

the influence of psychological factors brought by driving, such as stress, fatigued driving, etc. In addition, real road driving often encounters more complex environmental conditions, which are more difficult for drivers to make decisions. Therefore, real road experiment on driving behaviour is a process worth further studying.

5. Conclusion

In this paper, we conducted experimental research on the impact of the in-vehicle AR-HUD system on driving behaviour, comprehensively analysed the subjects' eye movement data and risky reaction time during driving, and obtained the independent variable with (or without) the AR-HUD system. The experimental results with the count of fixations in the AOI, the mean fixation time, fixation duration percentage, the risky perception time, and the risky decision-making time as the dependent variables, the main conclusions are listed as follows:

- (1) In the night driving situation, the AR-HUD system can significantly improve the subjects' attention to risky AOI, reduce the difficulty of processing information in risky driving scenarios, and reduce recognition load. In different risky driving situations,

the extent of the impact is significantly different accordingly. While in the daytime driving situation, the impact is not significant.

- (2) The AR-HUD system can significantly shorten the subjects' perception time of risky driving scenarios and respond to risky situations more quickly, but it has no significant impact on the risky decision-making time.
- (3) Subjective acceptance of the AR-HUD system was higher than expected after implementing the system; and the subjects' evaluation of the effectiveness, interference and reliability of the system was affected by driving experience and gender significantly.

CRediT authorship contribution statement

Yu-nuo Cheng: Conceptualization, Methodology, Validation, Writing – original draft. **Xia Zhong:** Data curation, Writing – review & editing, Supervision. **Li-wei Tian:** Visualization, Investigation, Software, Resources.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Cheng Yunuo reports financial support was provided by China University of Mining and Technology. Tian Liwei reports a relationship with Chongqing Changan Automobile Co Ltd that includes: non-financial support.

Data availability

The authors do not have permission to share data.

Acknowledgement

This paper was funded by the Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX21_2128). We also thank Changan Automobile Group Co., Ltd for assistance with the experiment.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trip.2023.100767>.

References

- Bark, K., Tran, C., Fujimura, K., et al., 2015. Personal Navi: Benefits of an augmented reality navigational aid using a see-Thru 3D volumetric HUD. In: *International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. ACM, pp. 1–8.
- Bellazzi, A., Bellia, L., Chinazzo, G., et al., 2022. Virtual reality for assessing visual quality and lighting perception: A systematic review. *Build. Environ.* 209.
- Castro, S.C., Strayer, D.L., Matzke, D., Heathcote, A., 2019. Cognitive workload measurement and modelling under divided attention. *Journal Of Experimental Psychology-Human Perception And Performance* 45 (6), 826–839. <https://doi.org/10.1037/xhp0000638>.
- Choi, S.Y., Kim, M.Y., Joe, H.I., 2002. A Study on the Performance Index and visibility for LED Variable Message Sign. *J. Korean Instit. Electric. Eng. Summer Conf.* 10–12.
- Choi, W.J., Lee, W.J., Lee, S.H., et al., 2013. Vehicle HUD's cognitive emotional evaluation-focused on colour visibility of driving information. *Korean J. Sci. Emotion Sensibility* 195–206.
- Christopher, J., Patten, D., Kircher, Albert, Ostlund, Joakim, Nilsson, Lena, 2004. Using mobile telephones: cognitive workload and attention resource allocation. *Acc. Anal. Prevent.* 36 (3), 341–350. [https://doi.org/10.1016/S0001-4575\(03\)00014-9](https://doi.org/10.1016/S0001-4575(03)00014-9).
- Cooper, J.M., Medeiros-Ward, N., Strayer, D.L., 2013. The impact of eye movements and cognitive workload on lateral position variability in driving. *Hum. Factors* 55 (5), 1001–1014. <https://doi.org/10.1177/0018720813480177>.
- Dong, Y.C., Hu, Z.C., Uchimura, K., Murayama, N., 2011. Driver inattention monitoring system for intelligent vehicles: a review. *IEEE Trans. Intell. Transport. Syst.* 12 (2), 596–614. <https://doi.org/10.1109/TITS.2010.2092770>.
- Fan, W.u., Liang, Z., 2012. Review of augmented reality technology development and application. *Comput. Knowl. Technol.* 34, 8319–8325.
- Fröhlich, P., et al., 2010. Augmenting the Driver's View with Realtime Safety-Related Information. *Augmented Human Int. Conf., Megève, France*, pp. 1–11.
- Gibbered, D.L., Jameson, S.L., Carsten, O.M.J., 2010. Managing in-vehicle distraction: evidence from the psychological refractory period paradigm. *The 2nd International Conference on Automotive User Interface and Interactive Vehicular Applications, Pittsburgh*.
- Grandi, Fabio, Prati, Elisa, Peruzzini, Margherita, 2022. Design of ergonomic dashboards for tractors and trucks: innovative method and tools. *J. Indust. Inform. Integrat.* 25, 100304. <https://doi.org/10.1016/j.jii.2021.100304>. ISSN 2452-414X.
- Hao, T., Zhengyu, T., Chunhui, J., 2015. *Design of Human-Computer Interface*. Electronic Industry Press, Beijing, pp. 6–9.
- Huawei, Z., 2015. Research on HUD visual interface design of vehicle head-up display [D]. South China University of Technology, Guangzhou.
- Hwang, Y., Park, B.J., Kim, K.H., 2015. The effects of augmented-reality head-up display system on the perception of precautionary situation. In: *International Conference on Information and Communication Technology Convergence*. IEEE, pp. 1146–1148.
- Hwang, Y., Park, B.J., Kim, K.H., 2016. Effects of augmented-reality head-up display system use on risk perception and psychological changes of drivers. *ETRI J.* 757–766.
- Jenness, J.W., Singer, J., 2008. Visibility and conspicuity of detectable warnings for pedestrians with visual impairments. *Transport. Res. Rec.* 104–113, 2073.
- Kim, K., Hwang, Y., 2017. The usefulness of augmenting reality on vehicle head-up display. *Adv. Human Aspects Transport.* 25–36.
- Kim, H., Miranda, A., Misu, T., et al., 2016. Look at me: augmented reality pedestrian warning system using an in-vehicle volumetric head-up display. In: *International Conference on Intelligent User Interfaces*. ACM, pp. 294–298.
- Lee, S.H., Choi, J., Park, J., 2009. Interactive e-learning system using pattern recognition and augmented reality. *IEEE Transactions. Consum. Electron.* 55 (2), 883–890.
- Lee, J.W., Park, B.J., Yoon, C.R., et al., 2015. A study of issues and considerations for development of a vehicle AR system. In: *International Conference on Information and Communication Technology Convergence*, pp. 1160–1162.
- Lehtonen, E., Lappi, O., Koirikivi, I., Summala, H., 2014. Effect of driving experience on anticipatory look-ahead fixations in real curve driving. *Acc. Anal. Prevent.* 70, 195–208. <https://doi.org/10.1016/j.aap.2014.04.002>.
- Liang, Y.L., Reyes, Michelle L., Lee, John D., 2007. Real-time detection of driver cognitive distraction using support vector machines. *IEEE Trans. Intell. Transport. Syst.* 8 (2), 340–350. <https://doi.org/10.1109/TITS.2007.895298>.
- Liu, Y., Wen, M., 2008. Comparison of head-up display (HUD) vs. head-down display (HDD): driving performance of commercial vehicle operators in Taiwan. *Int. J. Hum. Compute. Stud* 61 (5), 679–697.
- Liu, N.C., Xie, L., Wu, C.Z., Fu, Q., Deng, C., 2017. Driver's cognitive workload and driving performance under traffic sign information exposure in complex environments: A case study of the highways in China. *Internat. J. Environ. Res. Public Health* 14 (2). <https://doi.org/10.3390/ijerph14020203>.
- Ma, R.Q., Kaber, D.B., 2005. Situation awareness and workload in driving while using adaptive cruise control and a cell phone. *Internat. J. Indust. Ergon.* 35 (10), 939–953. <https://doi.org/10.1016/j.jergon.2005.04.002>.
- Ma, Y.L., Qi, S.M., Zhang, Y.P., Lian, G., Lu, W.X., Chan, C.Y., 2020. Drivers' visual attention characteristics under different cognitive workloads: an on-road driving behaviour study. *Internat. J. Environ. Res. Public Health* 17 (15). <https://doi.org/10.3390/ijerph17155366>.
- Mark, A., Wetton, A.H., 2011. The development and validation of a hazard perception test for use in driver licensing. *Accid. Anal. Prev.* 43 (5).
- Medenica, Z., et al., 2011. Augmented reality vs. street views: A driving simulator study comparing two emerging navigation aids. In: *Proc. Int. Conf. HCI Mobile Devices Services, Stockholm, Sweden*, pp. 265–274.
- Moffitt, K., Browne, M.P., 2019. Visibility of colour symbology in head-up and head-mounted displays in daylight environments. *Opt. Eng.* 58 (5). <https://doi.org/10.1117/1.OE.58.5.051809>.
- NHTSA. Traffic Safety Facts: Distracted Driving 2010[R]. NHTSA's National Centre for Statistics and Analysis, Washington, 2012: 113–120.
- Park, H.S., Kim, K.H., 2014. AR-Based Vehicular Safety Information System for Forward Collision Warning. In: *Virtual, Augmented and Mixed Reality. Applications of Virtual and Augmented Reality*. Springer International Publishing, pp. 435–442.
- Park, B.J., Lee, J.W., Yoon, C., Kim, K.H., 2015. Augmented reality and representation in vehicle for safe driving at night. In: *2015 Internat. Conf. ICT Converg. (ICTC)*, pp. 1261–1263.
- Smith, S., Fu, S.H., 2011. The relationships between automobile head-up display presentation images and drivers' Kansei. *Displays* 32 (2), 58–68. <https://doi.org/10.1016/j.displa.2010.12.001>.
- Smith, M., Gabbard, J.L., Burnett, G., et al., 2017. The effects of augmented reality head-up displays on drivers' eye scan patterns, performance, and perceptions. *Internat. J. Mobile Human-Comput. Interact.*
- The SPSSAU project (2022). SPSSAU. (Version 22.0) [Online Application Software]. Retrieved from <https://www.spssau.com>.
- R. Venkatasawmy, 'The Digitization of Cinematic Visual Effects: Hollywood's Coming of Age'. Washington, DC, USA: Lexington Books, 2012.[F. Schewe, M. Vollrath, "Ecological interface design effectively reduces cognitive workload – The example of HMI for speed control," *Transportation Research Part F: Traffic Psychology and Behaviour*. Vol.72, 2020, pp.155-170.
- Venkatesh, Morris, M.G., Davis, G.B., 2003. User acceptance of information technology: Toward a unified view. *MIS Q.* 27 (3), 425–478. <https://doi.org/10.2307/30036540>.
- Wang, Z.S., Li, J., Feng, C.K., et al., 2016. A new method to evaluate the readability of head-up display. In: *2016 2nd IEEE International Conference On Computer And Communications*, pp. 1019–1023.
- Ward, N.J., Parkes, A., 1994. 'Head-up displays and their automotive application: An overview of human factors issues affecting safety'. *Accident Anal. Prevention* 26 (6), 703–717.
- Ward, N.J., Parkes, A., Crone, P.R., 1995. Effect of background scene complexity and field dependence on the legibility of head-up displays for automotive applications. *Hum. Factors* 37 (4). <https://doi.org/10.1518/001872095778995625>.
- Xing, W., Qi, Q., 2014. Vehicle head-up display technology. *Electro-Optics Control* 01, 55–58.
- Yamaguchi, K; Nakayama, M; Sun, Q; Xia, JH. Prediction of Cognitive Performance of Drivers using Eye Fixation Behaviours. 2019 23RD International Conference Information Visualisation (IV): Biomedical Visualization And Geometric Modelling & Imaging. 2019. 186-190:10.1109/IV.2019.00040.
- Yoon, C., Kim, K., Park, H.S., et al., 2014. Development of augmented forward collision warning system for head-up display. In: *Proceedings of 17th International IEEE Conference on Intelligent Transportation Systems*, p. 3.
- Zhang, Y.J., Yang, T., Zhang, X., Zhang, Y.J., Sun, Y.C., 2021. Effects of full windshield head-up display on visual attention allocation. *Ergonomics* 64 (10), 1310–1321. <https://doi.org/10.1080/00140139.2021.1912398>.